

NIKOS DOKMETZOGLU

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EDUCATION

University of North Carolina at Chapel Hill

Chapel Hill, NC, USA

Doctor of Philosophy (PhD) in Physics

August 2023

- Dissertation Title: “[Conformal Yangian and Tree Amplitudes in Scalar and Gauge Field Theories](#)”
- PhD Advisor: Prof. Louise Ann Dolan

Master of Science (MSc) in Physics

December 2019

Davidson College

Davidson, NC, USA

Bachelor of Science (BSc) in Physics

May 2017

- Magna Cum Laude, Honors in Physics, Minor in Mathematics
- Cumulative GPA: 3.96 Major GPA: 3.98 Minor GPA: 4.00

Athens College (Hellenic-American Educational Foundation)

Athens, Greece

Highschool Diploma, Salutatorian 2013

July 2013

ADDITIONAL TRAINING

Harvard University (HarvardX)

Online

CS50's Introduction to Artificial Intelligence with Python [🌐]

Spring 2025

CS50's Introduction to Programming with Python [🌐]

Fall 2024

Wolfram U

Online

Exploring AI Foundations with Wolfram Tools [🌐]

January 2025

1. Introduction to Machine Learning in Wolfram Language [🌐]
2. Introduction to Neural Networks in Wolfram Language [🌐]
3. Wolfram Language and LLMs [🌐]

Parallel Programming in Wolfram Language [🌐]

November 2024

Advancing Physics Research and Teaching with Wolfram Language [🌐] [📺]

October 2024

Wolfram Language Programming Proficiency [🌐]

October 2024

1. Wolfram Language Programming Fundamentals [🌐] [📺]
2. Practical Programming with Wolfram Language [🌐] [📺]
3. Programming and Development in Wolfram Language [🌐] [📺]

Writing High-Performance Code [🌐] [📺]

October 2024

Introduction to Machine Learning in Wolfram Language [🌐] [📺]

September 2023

Deutsches Elektronen-Synchrotron (DESY) Hamburg

Online

Computer Algebra and Particle Physics (CAPP) 2023 [🌐]

July 2023

Charles University

Prague, Czech Republic

Amplitudes 2022 Summer School [🌐]

August 2022

Mainz Institute for Theoretical Physics MITP School 2021: The Amplitude Games [🌐]	Online July 2021
Institut de Physique Théorique (IPhT), CEA/CNRS-Saclay SAGEX Mathematica & Maple School [🌐]	Online January 2021
Brown University Amplitudes 2020 (Zoomplitudes) Master Class [🌐]	Online May 2020
Instituto de Física Teórica (IFT) UAM-CSIC Summer IFT School (SIFTS) 2019 [🌐]	Madrid, Spain July 2019
Perimeter Institute for Theoretical Physics Tri-Institute Summer School on Elementary Particles (TRISEP) 2018 [🌐]	Waterloo, ON, Canada July 2018
University of California, Davis, QMAP Amplitudes 2018 Summer School [🌐]	Davis, CA, USA June 2018

RESEARCH EXPERIENCE [🏠]

National Centre for Scientific Research “Demokritos” <i>Postdoctoral Researcher</i>	Athens, Greece Fall 2023 – Fall 2024
■ “Analytic Continuation of Five-Point Two-Loop Master Integrals” [In Progress] ■ Joined the efforts of the research group of Prof. Costas Papadopoulos, at the Institute of Nuclear and Particle Physics of the National Centre for Scientific Research “Demokritos”, to analytically express all five-point two-loop master integrals with one off-shell leg and to provide computational tools for their efficient numerical evaluation, using the so-called simplified differential equations approach.	
Max Planck Institute for Physics <i>Visiting Researcher</i>	Munich, Germany Spring 2023
■ “Remnants of Conformal Symmetry at Loop Level” [In Progress] ■ Joined the quantum field theory and scattering amplitudes research group at the Max Planck Institute for Physics in Munich, led by Prof. Johannes Henn, to investigate the remnants of conformal symmetry in loop-level scattering amplitudes, with the goal of better understanding the structure of the algebraic coefficients appearing in the decomposition of on-shell loop amplitudes into a basis of scalar integrals.	
University of North Carolina at Chapel Hill <i>Doctoral Dissertation Research</i>	Chapel Hill, NC, USA Fall 2018 – Summer 2023
■ “Conformal Yangian and Tree Amplitudes in Scalar and Gauge Field Theories” ■ Conducted research in quantum field theory under the guidance of Prof. Louise Ann Dolan. ■ Proved the algebraic consistency of the conformal Yangian $Y[\mathrm{SO}(2, n)]$, i.e. the infinite-dimensional Yangian extension of the conformal group $\mathrm{SO}(2, n)$, where n is the number of space-time dimensions, by showing that the momentum-space differential operator representation of its generators satisfies the so-called Serre relation, for both scalar and spin-one gauge fields. ■ Investigated the action of the $Y[\mathrm{SO}(2, n)]$ generators on the tree-level scattering amplitudes of scalar $\lambda\varphi^3$ theory and pure Yang-Mills theory, two non-supersymmetric field theories which are connected through the Cachazo-He-Yuan (CHY) formalism and known to be conformally invariant at tree-level in $n = 6$ and $n = 4$ space-time dimensions, respectively. Examined the action of the $Y[\mathrm{SO}(2, n)]$ generators on the off-shell scattering polynomials appearing in the polynomial form of the CHY formalism.	

Davidson College

Merzbacher Summer Research Fellow

Honors Thesis in Physics

Weinstein Davidson Research Initiative Summer Research Fellow

- “Implementation of Recursion Relations in Gluon Scattering Amplitude Calculations in $\text{AdS}_4/\text{CFT}_3$ ”
- Conducted research in **quantum field theory** under the guidance of **Prof. Savan Kharel**.
- Used generalized Britto–Cachazo–Feng–Witten (BCFW) **recursion relations** and the spinor-helicity formalism to compute four-point and five-point **tree-level gluon scattering amplitudes in $\text{AdS}_4/\text{CFT}_3$** (Anti-de Sitter/Conformal Field Theory). Used the symbolic manipulation system FORM and Mathematica to simplify amplitude calculations.

Davidson, NC, USA

Summer 2018

Fall 2016 – Spring 2017

Summer 2016

Davidson College

Independent Research in Physics

Davidson College Faculty Study and Research Grant Summer Research Fellow

- “Momentum Tails of 1D, 2D, and 3D Quantum Systems”
- Conducted research in **quantum mechanics** under the guidance of **Prof. Mario Belloni**.
- Utilized Mathematica and its parallel processing capabilities to study the **large-momentum behavior** of the **momentum-space wavefunctions** of different quantum systems. Discovered a dependence of the large-momentum $1/P^\beta$ tail of the momentum-space wavefunction on the dimensionality and the potential energy function of a given quantum system.

Davidson, NC, USA

Spring 2016

Summer 2015

PUBLICATIONS [IN]

N. Dokmetzoglou and L. Dolan, *Properties of the Conformal Yangian in Scalar and Gauge Field Theories*, *JHEP* **02** (2023) 137 [[arXiv:2207.14806 \[hep-th\]](#)].

N. Dokmetzoglou, CONFORMALYANGIAN: *a MATHEMATICA Package for Computations Related to the Action of the Conformal Yangian $Y[\text{SO}(2, n)]$* , in preparation.

PRESENTATIONS

“Analytic Continuation of Five-Point Two-Loop Master Integrals”

- [NCSR-D INPP – APCTP Meeting & HOCTools-II Mini-Workshop](#), Athens, Greece, October 2024
- [Amplitudes 2024 Gong Show](#), Princeton, NJ, USA, June 2024

“Conformal Yangian and Tree Amplitudes in Scalar and Gauge Field Theories”

- [NCSR Demokritos INPP Seminar](#), Online, June 2023
- [Amplitudes 2022 Summer School Gong Show](#), Prague, Czech Republic, August 2022

“Implementation of Recursion Relations in Gluon Scattering Amplitude Calculations in $\text{AdS}_4/\text{CFT}_3$ ”

- [APS April Meeting 2017](#), Washington, DC, USA, January 2017
- [Quadrennial Physics Congress \(PhysCon\) 2016](#), Silicon Valley, CA, USA, November 2016

TEACHING EXPERIENCE [---

University of North Carolina at Chapel Hill

Graduate Teaching Assistant

Chapel Hill, NC, USA

Fall 2017 – Spring 2023

Served as a teaching assistant for the lab component of the following introductory undergraduate courses. These courses were taught in the [lecture/studio](#) format.

- [PHYS 118](#) – Introductory Calculus-based Mechanics and Relativity *5 semesters*
- [PHYS 119](#) – Introductory Calculus-based Electromagnetism and Quanta *7 semesters*
- [PHYS 115](#) – General Physics II: For Students of the Life Sciences *1 semester*

Led weekly problem-solving sessions for the following graduate-level core courses.

- [PHYS 701](#) – (Graduate) Classical Dynamics [] [] [] *3 semesters*
- [PHYS 712](#) – (Graduate) Electromagnetic Theory [] [] [] *3 semesters*

Davidson College

Physics Tutor, Center for Teaching and Learning

Davidson, NC, USA

Fall 2015 – Spring 2017

LEADERSHIP EXPERIENCE

Society of Physics Students (National)

Zone 5 (NC and SC) Associate Zone Councilor

Davidson, NC, USA

June 2016 – June 2017

Served as a link between Zone 5 chapters and the national office. Coordinated zone activities and represented North Carolina and South Carolina students at the SPS National Council. Served as a member of the Professional Development Committee and participated in the organization of [PhysCon 2016](#).

Society of Physics Students (Davidson College Chapter)

President

Davidson, NC, USA

March 2016 – March 2017

Organized and presided over meetings of the Davidson College SPS chapter. Was responsible for the chapter's finances and overall operation. Composed the [SPS Chapter Report 2015-2016](#), which resulted in our chapter's sixth consecutive Outstanding Chapter Award.

Vice-President of Professional Affairs

March 2015 – March 2016

Maintained the chapter's connection with the professional community through zone and national SPS meetings. Organized Physics GRE preparation sessions taught by the department's faculty. Launched a mentoring program that connected upperclass and younger physics majors.

HONORS / AWARDS

- Hamilton and Silver Awards, UNC Chapel Hill, 2023
- Hamilton and Silver Awards, UNC Chapel Hill, 2022
- Hamilton Award, UNC Chapel Hill, 2021
- [Outstanding Graduate Teaching Assistant Award](#), UNC Chapel Hill, 2020
- Merzbacher Fellowship, UNC Chapel Hill, 2018
- [Honors in Physics](#), Davidson College, 2017
- Physics Award, Davidson College, 2017
- [Theoretical/Computational Physics Poster Award](#), PhysCon 2016

HONOR SOCIETIES / PROFESSIONAL MEMBERSHIPS

- Member of: [ΦBK](#) (2017), [ΣΠΣ](#) (2016), [Bernard Society of Mathematics](#) (2015)
- Member of: [American Physical Society](#) (2013), [Society of Physics Students](#) (2013)

PARTICIPATION IN CONFERENCES, WORKSHOPS AND MEETINGS

- [Xmas Theoretical Physics Workshop @Athens 2024](#), Athens, Greece, December 2024
- [YOUNGST@RS – Loop-the-Loop: Feynman Calculus and its Applications to Gravity and Particle Physics](#), Online, November 2024
- [NCSR-D INPP – APCTP Meeting & HOCTools-II Mini-Workshop](#), Athens, Greece, October 2024
- [Amplitudes 2024 Conference](#), IAS, Princeton, NJ, USA, June 2024
- [Current Topics in Fundamental Physics](#), Monastery Benediktbeuern, Germany, August 2023
- [Amplitudes 2023 Conference](#), CERN, Meyrin, Switzerland, August 2023
- [Xmas Theoretical Physics Workshop @Athens 2022](#), Athens, Greece, December 2022
- [QCD Meets Gravity 2022](#), Zurich, Switzerland, December 2022
- [Amplitudes 2022 Conference](#), Prague, Czech Republic, August 2022
- [Amplitudes 2021 Conference](#), Online, August 2021
- [Strings 2021 Conference](#), Online, June 2021
- [New Trends on Quantum Field Theory, Amplitudes and Gravity](#), Online, March 2021
- [Iberian Strings 2021](#), Online, January 2021
- [Strings 2020 Conference](#), Online, June 2020
- [Amplitudes 2020 \(Zoomplitudes\) Conference](#), Online, May 2020
- [APS April Meeting 2020](#), Online, April 2020
- [APS April Meeting 2017](#), Washington, DC, USA, January 2017
- [Quadrennial Physics Congress \(PhysCon\) 2016](#), Silicon Valley, CA, USA, November 2016

SKILLS

- **Programming Languages:** Wolfram Language, Python, Java, Arduino
- **Scientific Computing & Computer Algebra Systems:** Wolfram Mathematica, Maple, FORM, NI LabVIEW, NI Multisim, Vernier Logger Pro, Easy Java Simulations, GeoGebra
- **System Administration & Operating Systems:** Windows, Linux, Ubuntu, WSL
- **Software Development & Version Control:** Visual Studio Code, GitHub, Git
- **Generative Artificial Intelligence Tools:** LM Studio, Ollama, AnythingLLM
- **Document Typesetting & Office Productivity Tools:** $\text{\LaTeX}$, Microsoft Office, Adobe Acrobat Pro, Google Docs Editors, Zotero
- **Video Conferencing Management & Recording:** Zoom, Microsoft Teams, Skype, Cisco Webex
- **Video & Music Production:** OBS Studio, Steinberg Cubase
- **Web Content Management:** WordPress, YouTube Studio
- **Spoken Languages:** English (fluent), Greek (native), Spanish (pre-intermediate), German (beginner)

REFERENCES

Prof. Louise A. Dolan

PhD Thesis Advisor

Kenan Distinguished Professor

Department of Physics and Astronomy

University of North Carolina at Chapel Hill

ldolan@physics.unc.edu

Prof. Costas G. Papadopoulos

Postdoc Supervisor

Director of Research

Institute of Nuclear and Particle Physics

N.C.S.R. “Demokritos”

cgppdo@inp.demokritos.gr

Prof. Johannes M. Henn

Research Visit Supervisor

Director

Department of Quantum Field Theory

Max Planck Institute for Physics

henn@mpp.mpg.de

Prof. Duane L. Deardorff

Graduate Teaching Assistantship Supervisor

Director of Undergraduate Laboratories

Department of Physics and Astronomy

University of North Carolina at Chapel Hill

duane.deardorff@unc.edu

Prof. Savan Kharel

BSc Thesis Advisor

Assistant Instructional Professor

Department of Physics

University of Chicago

skharel@uchicago.edu

Prof. Mario Belloni

BSc Thesis Advisor

Professor & Chair

Department of Physics

Davidson College

mabelloni@davidson.edu